

Monte Carlo Current and Proposed

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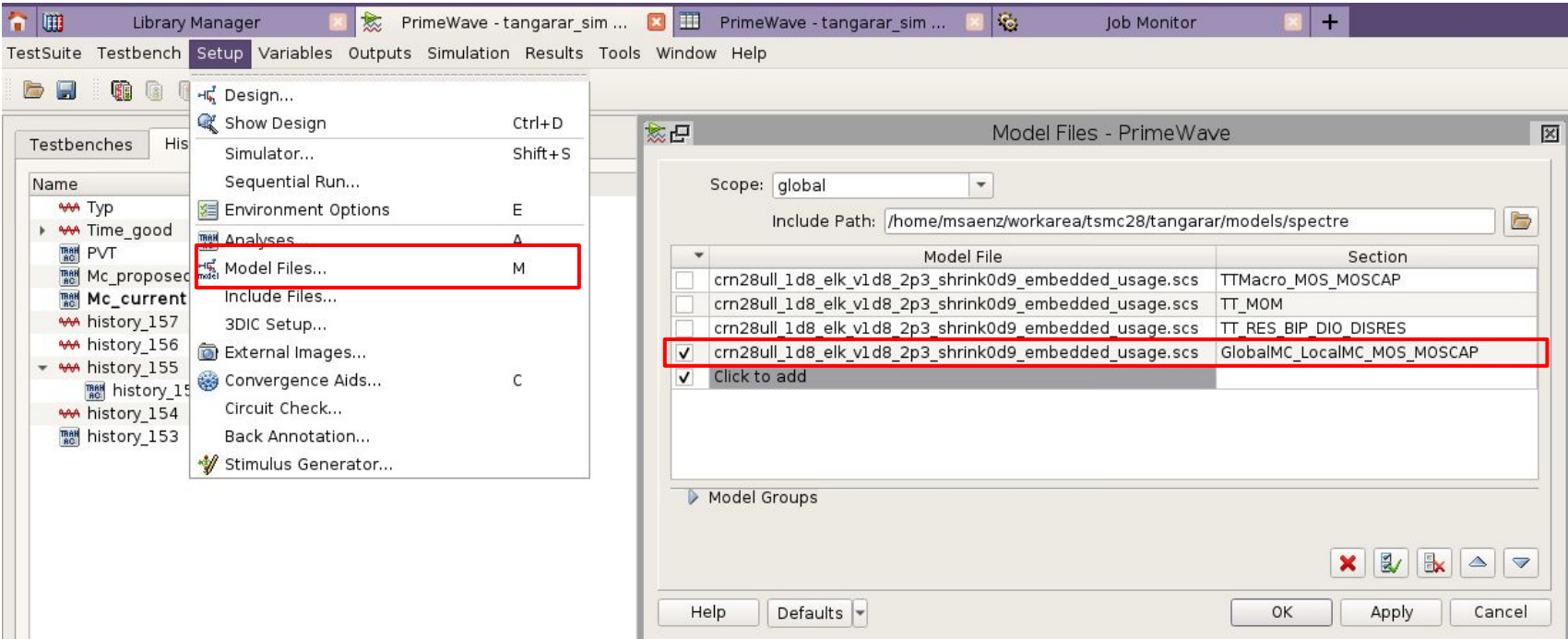
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Santander



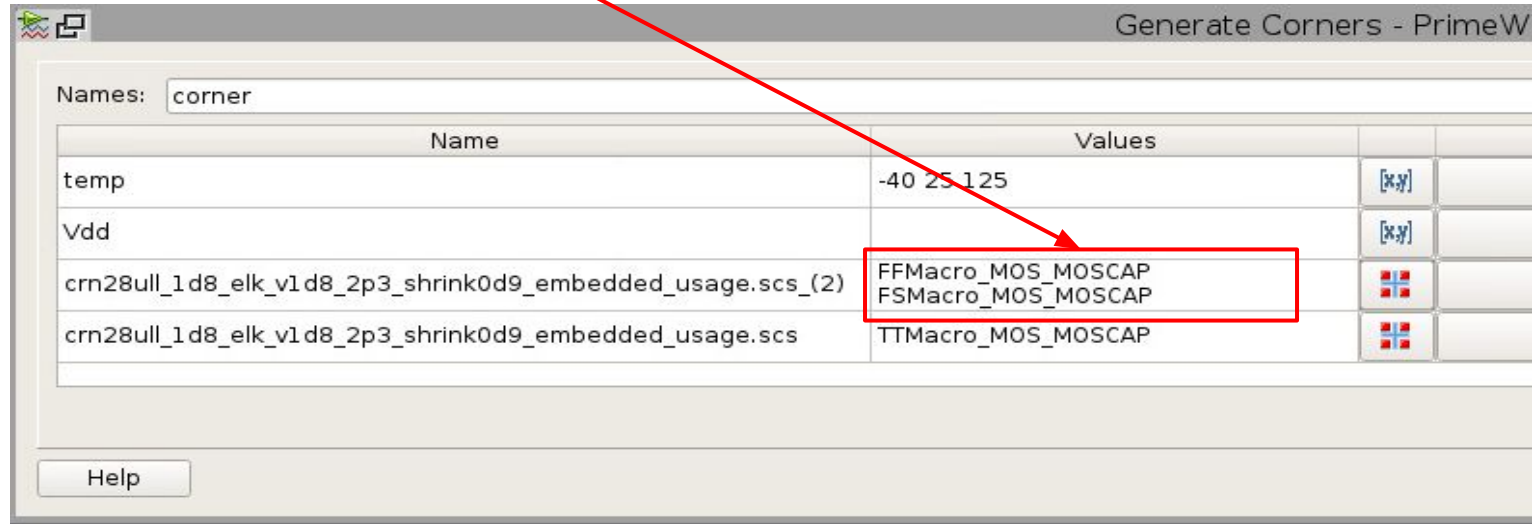
Monte Carlo

- Include models



Current - Monte Carlo

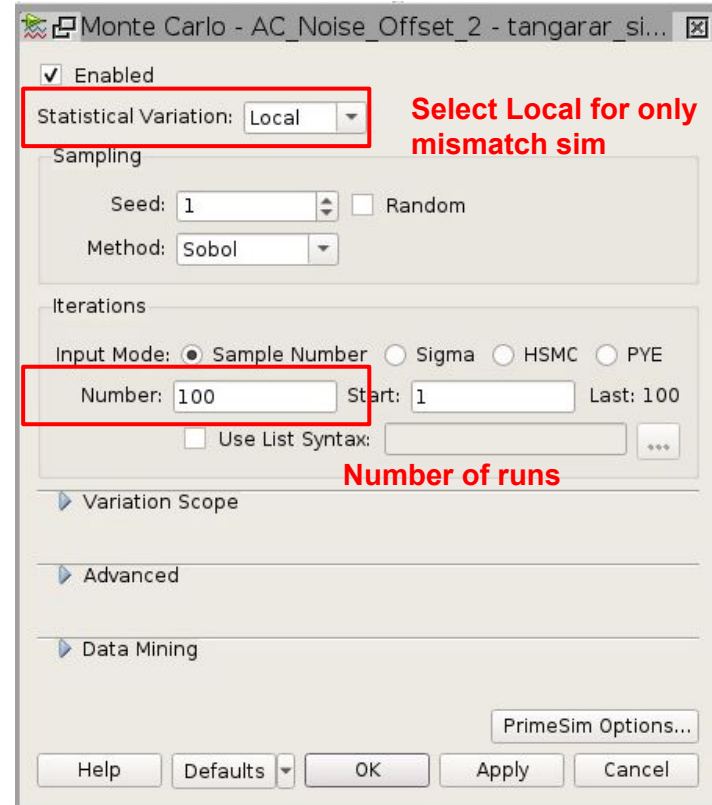
- Include the following models for corners (PVT):
 - FFMacro_MOS_MOSCAP
 - FSMacro_MOS_MOSCAP
 - SFMacro_MOS_MOSCAP
 - SSMacro_MOS_MOSCAP



Current - Monte Carlo

- MC run at Typ only.
- Use **only Corner sim**, then **only mismatch sim**.

PVT_{max(min)} +/- 1.133*(3*σ).



Current - Monte Carlo

PVTmin Ao

Corner		Corner Variable temp		Corner Variable Vdd		Corner Variable _2p3_shrink0d9		Ao_db acMeasure	
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
corner31				1.6				71.849	
worstCases_corne...						FFMacro_MO...		71.849	
corner36				1.8				72.649	
corner41				2				72.984	
corner32				1.6				73.732	
corner37				1.8		FSMacro_MO...		74.578	

PVTmax Ao

Corner		Corner Variable temp		Corner Variable Vdd		Corner Variable _2p3_shrink0d9		Ao_db acMeasure	
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
corner14				2				84.419	
worstCases_corne...								84.419	
corner9				1.8		SSMacro_MO...		83.962	
corner4				1.6				82.536	

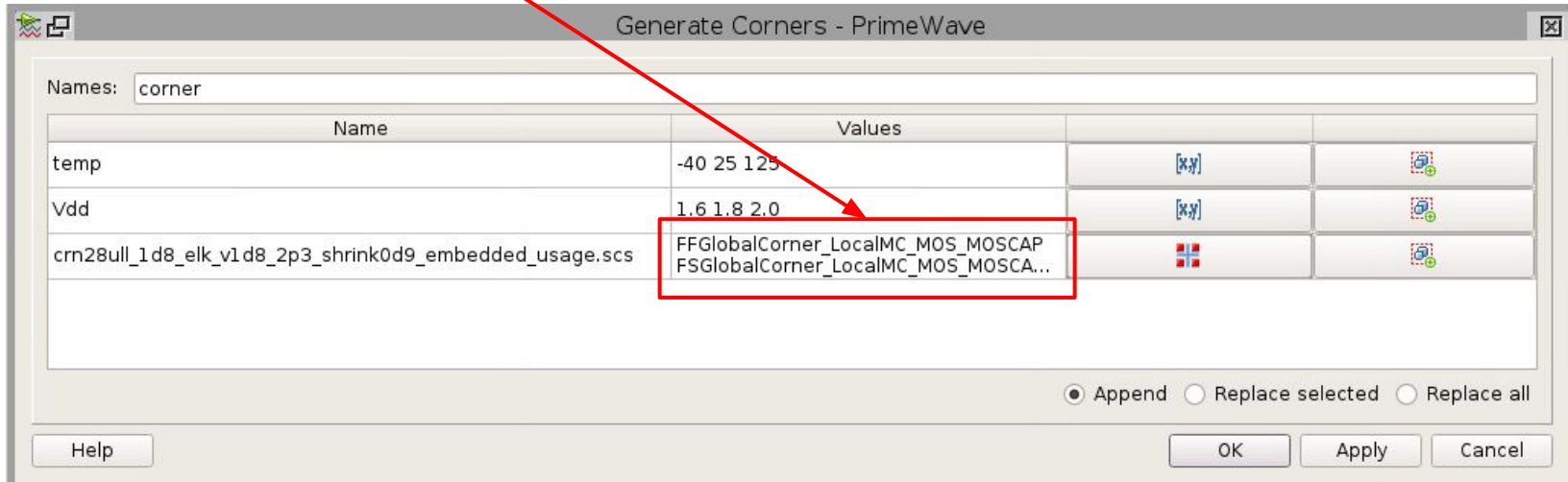
Monte Carlo simulation

Equations		Specification		Violations		Yield		Passed		Min		Max		Mean		StdDev		Median	
Filter	Filter	Filter	Filter	< Filter	< Filter	< Filter	< Filter	< Filter	< Filter	< Filter	< Filter	< Filter	< Filter	< Filter	< Filter	< Filter	< Filter	< Filter	< Filter
Ao_db:acMeasure		>70		15		85		85		63.696		108.2		79.6449		10.3855		77.7485	

σ

Proposed - Monte Carlo

- Include the following models for corners (PVT):
 - FFGlobalCorner_LocalMC_MOS_MOSCAP
 - FSGlobalCorner_LocalMC_MOS_MOSCAP
 - SFGlobalCorner_LocalMC_MOS_MOSCAP
 - SSGlobalCorner_LocalMC_MOS_MOSCAP



Proposed - Monte Carlo

PVTworst Ao

Corner		Corner Variable temp		Corner Variable Vdd		Corner Variable _2p3_shrink0d9		Ao_db acMeasure	
Filter	▼	Filter	▼	Filter	▼	Filter	▼	Filter	▼
corner151				1.6				71.883	
corner176				1.8		FFGlobalCor...		72.683	
corner201				2				73.018	
corner156						FSGlobalCor...		74.415	
corner171		125		1.6		TTMacro_MO...		74.978	

PVTbest Ao

Corner		Corner Variable temp		Corner Variable Vdd		Corner Variable _2p3_shrink0d9		Ao_db acMeasure	
Filter	▼	Filter	▼	Filter	▼	Filter	▼	Filter	▼
corner66				2				84.396	
corner41		-40		1.8				83.941	
corner16				1.6		SSGlobalCor...		82.526	
corner141		25						82.359	
corner61				2		SFGlobalCor...		82.211	

Worst Cases Global Corners - PrimeWave

Scope: global

Corners (enabled / total): 2 / 2

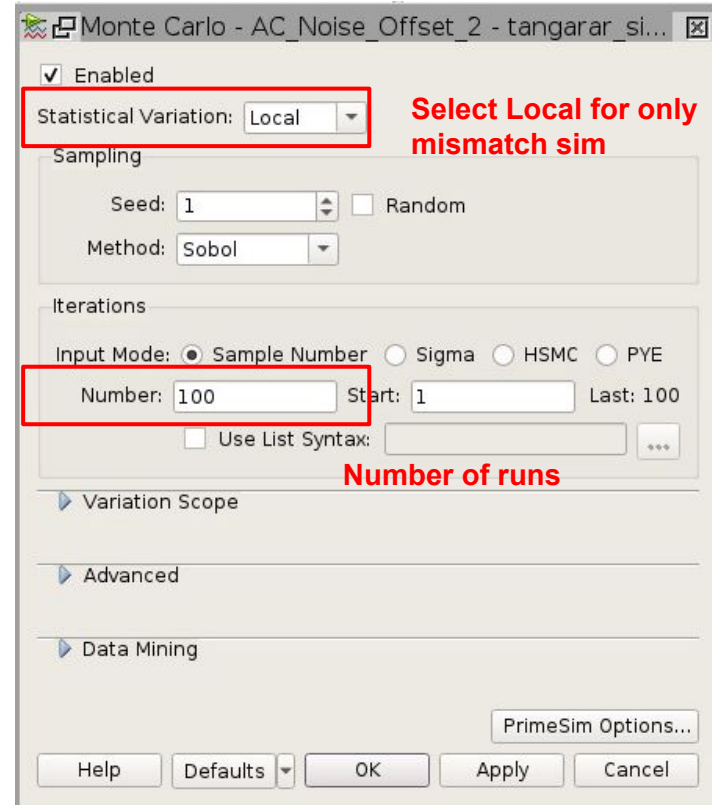
Temperature	☒ worst_Ao	☒ Best_Ao
☒ temp	125	-40
Design Variables		
☒ vdd	1.6	2
Model Files		
☒ crn28ull_1d8_elk_v1d8_2p3_shrink0d9_embedded_usage.scs	FFGlobalCor...	SSGlobalCor...
Testbenches		
☒ DC_performance_active		
☒ Time_response		
☒ AC_response		
☒ OP		
☒ Time_PD		
☒ Noise		
☒ Offset		

☒ Show measure names

HelpOKApplyCancel

Proposed - Monte Carlo

- MC should simulate at each PVT worst corner condition.
- Use **only Corner sim, then only mismatch sim.**
- Max/Min report: **meanPVT(worst | best) + 1.133*(3*sigma)**. This is obtained with the Global Corner + Local MC setup explained before.



Proposed - Monte Carlo

Equations		Specification		Violations		Yield		Passed		Min		Max		Mean		StdDev		Median	
Filter	▼	Filter	▼	< Filter	▼	< Filter	▼	< Filter	▼	< Filter	▼	< Filter	▼	< Filter	▼	< Filter	▼	< Filter	▼
Ao_db:acMeasure		>70		38		62		62		61.969		98.323		73.142		7.16036		71.698	
Ao_db:acMeasure		>70		8		92		92		64.377		130.62		81.3882		12.0129		79.6535	

81.3882
Spec: >70
corner=Best_Ao
temp=-40
Vdd=2
crn28ull_1d8_elk_v1d8_2p3_shrink0d9_embedded_usage.scs=SSGlobalCorner_LocalMC_MOS_MOSCAP

Monte Carlo Results

	MC_current		MC_proposed	
	Min	Max	Min	Max
Ao [dB]	36.5486	119.779	48.8039	122.22

Comparison of the two models

PVTmin Ao

Corner	Corner Variable temp	Corner Variable Vdd	Corner Variable _2p3_shrink0d9	Ao_db acMeasure
Filter	Filter	Filter	Filter	Filter
corner31	125	1.6	FFMacro_MO...	71.849
worstCases_corne...				71.849
corner36		1.8		72.649
corner41		2	FSMacro_MO...	72.984
corner32		1.6		73.732
corner37		1.8		74.578

With FFMacro_MOS_MOSCAP model

PVTworst Ao

Corner	Corner Variable temp	Corner Variable Vdd	Corner Variable _2p3_shrink0d9	Ao_db acMeasure
Filter	Filter	Filter	Filter	Filter
corner151	125	1.6	FFGlobalCor...	71.883
corner176		1.8		72.683
corner201		2		73.018
corner156		1.6	FSGlobalCor...	74.415
corner171			TTMacro_MO...	74.978

With FFGlobalCorner_LocalMC_MOS_MOSCAP model

Thanks

Percentiles

Muestras	Sigma	Percetil	75% confidence	95% confidence
100		P5	X3 a X98	X2 a X99
	1 σ	P15,87 P84,13	X13 a X88	X10 a X91
	2 σ	P2,275 P97,725	X1 a X100	No hay suficientes datos para informar de un intervalo
200		P5	X8 a X193	X5 a X196
	1 σ	P15,87 P84,13	X28 a X173	X23 a X178
	2 σ	P2,275 P97,725	X3 a X198	X1 a X200
300		P5	X12 a X289	X9 a X292
	1 σ	P15,87 P84,13	X34 a X258	X37 a X264
	2 σ	P2,275 P97,725	X5 a X296	X3 a X298
400		P5	X17 a X384	X13 a X388
	1 σ	P15,87 P84,13	X58 a X343	X52 a X349
	2 σ	P2,275 P97,725	X7 a X394	X5 a X396
500		P5	X22 a X479	X17 a X484
	1 σ	P15,87 P84,13	X74 a X427	X66 a X435
	2 σ	P2,275 P97,725	X9 a X492	X6 a X495
1400		P5	X64 a X1337	X57 a X1394
	1 σ	P15,87 P84,13	X213 a X1188	X200 a X1201
	2 σ	P2,275 P97,725	X28 a X1373	X23 a X1378